PROJECT · PCB DESIGN

Library creation and multi-sheet PCB in Altium Designer

Polytechnique Sousse · mini-project · April 2025

Multi-sheet
project with a top sheet

AD8630
quad op-amp used in the design

3D
bodies aligned on each footprint

CONTEXT

The goal was to turn a paper schematic into a manufacturable PCB, starting from an empty component library.

TOOLS AND METHODS

Altium Designer · PCB layout · Footprints · Schematic capture

WHAT I DID

- Created schematic symbols and footprints from datasheets, including a TO-220 package, a 1×10 screw terminal and a multi-turn trimmer.
- Added 3D bodies and aligned them with the footprints.
- Split the design into sheets: fan driver (TIP122), current reading and current regulation (AD8630).
- Generated the PCB, placed and routed it by hand, and prepared the fabrication layers.

OUTCOME

A reusable custom library and a complete multi-sheet PCB design.

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